

Implementing Gravitational Wave Emission in N-body Simulations: Applications to Hierarchical Triples

Mukhil Chandrakumar, Arun Kenath

Department of Physics and Electronics
CHRIST (Deemed to be University)

February 5, 2026

Gravitational Waves

- Einstein showed that there exist gravitational waves using his Theory of General Relativity (Einstein, 1918).
- GWs are perturbations on spacetime caused by asymmetry in the source, which allows for a quadrupole moment. Hence, the metric is given by

$$g_{\mu\nu} = n_{\mu\nu} + h_{\mu\nu} \quad (1)$$

- From the perturbed metric and Einstein's equation, we get the gravitational wave equation

$$\square h_{\mu\nu} = \frac{16\pi G}{c^4} T_{\mu\nu} \quad (2)$$

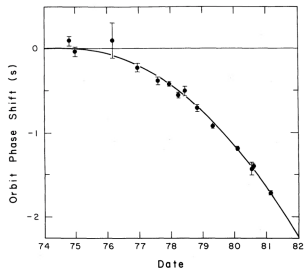


Figure 1: If energy loss by GW were to not occur, orbital period would have been constant over time. But observations indicate otherwise (Taylor and Weisberg, 1982)

Gravitational Waves

- In an eccentric orbit, the emission of gravitational waves is highly efficient and is distributed over many harmonics. The total radiation power $\mathcal{P}$ averaged over one period (Gupta et al., 2020) is given by

$$\mathcal{P} = \frac{32G^4 m_1^2 m_2^2 m_{\text{bin}}}{c^5 a_{\text{bin}}^5 (1 - e_{\text{bin}}^2)^{7/2}} \left(1 + \frac{73}{24} e_{\text{bin}}^2 + \frac{37}{96} e_{\text{bin}}^4 \right) \quad (3)$$

- It shows that the radiation power increases rapidly as the eccentricity increases.
- One such cause of creation of such eccentric binaries could be **Hierarchical Triple Systems (HTS)**.

Hierarchical Triple Systems

- Hierarchical Triple Systems (HTSs) consist of an inner and an outer binary.
- The stability of these systems (Gupta et al., 2020) is dependent on the condition beyond which there may be chaotic instability.

$$\frac{a_{\text{out}}}{a_{\text{in}}} > \frac{2.8}{1 - e_{\text{out}}} \left[\frac{m_{\text{out}}}{m_{\text{in}}} \frac{1 + e_{\text{out}}}{(1 - e_{\text{out}})^{\frac{1}{2}}} \right]^{\frac{2}{5}} \quad (4)$$

- From here on, we shall describe elements in the system using the subscripts 'in' and 'out' for the inner and outer binaries, respectively. Also, masses m_1 , m_2 form the inner binary and m_3 is the perturber.

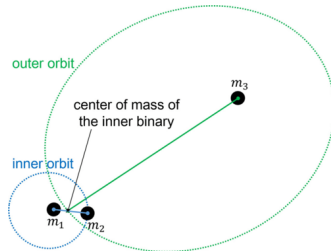


Figure 2: Simple Diagram of HTS
(Gupta et al., 2020)

Effects to Consider (Kozai-Lidov Mechanism)

- The Kozai-Lidov (KL) mechanism (Kozai, 1962; Lidov, 1962) is a dynamical phenomenon observable in such hierarchical triple systems.
- The relative inclination (I) and eccentricity (e_{in}) exchange is secular with respect to a conserved value Θ (Gupta et al., 2020),

$$\Theta = (1 - e_{\text{in}}^2) \cos^2 I \quad (5)$$

- Based on this, we find our range of initial inclination of the circular orbit for the KL oscillation (Gupta et al., 2020),

$$0 \leq |\cos I| \leq \sqrt{\frac{3}{5}} \quad \text{or} \quad 39.2315^\circ \lesssim I \lesssim 140.7685^\circ \quad (6)$$

Effects to Consider (General Relativistic Effects)

- The maximum eccentricity achieved by KL oscillations is restricted by the relativistic precession of the periastron/pericenter of the inner binary (Blaes et al., 2002). However, GR precession may still affect the system even if $t_{\text{KL}} < t_{\text{GR,in}}$ and can even retrigger KL mechanism in similar mass binaries (Naoz, 2016).
- The Lense-Thirring (LT) effect is a general relativistic correction to the precession of a gyroscope in the presence of a large rotating mass. This causes the relative orbital nodes of the gyroscope to precess (Pfister, 2007).

Background & Related Work

For a detailed analysis of hierarchical triple dynamics and the frequency-domain signatures of eccentric binaries, see our earlier work (preprint, article under peer-review):

M. Chandrakumar, A. Kenath, R. Sharma
Dynamical Evolution of Hierarchical Triple Systems and their Gravitational Wave Detectability
Preprints.org (2026)
DOI: 10.20944/preprints202601.1591.v1



Adding Radiation Reaction to REBOUND

- To observe GW emission from our inner binary, we add acceleration corrections of 2PN and 2.5PN order (Blanchet, 2014), to the accelerations of m_1 , m_2 simulated by the REBOUND + `gr_full`.

$$\begin{aligned}
 a_1^{2PN} = \frac{1}{c^4} \left[\left\{ -\frac{5G^2 m_1^2 m_2}{4r_{12}^4} - \frac{69G^3 m_1 m_2^2}{2r_{12}^3} - \frac{9G^3 m_2^3}{r_{12}^4} + \frac{Gm_2}{r_{12}^2} \left(-\frac{15}{8}(n_{12}v_2)^4 + \frac{3}{2}(n_{12}v_2)^2 v_1^2 - 6(n_{12}v_2)^2 (v_1 v_2) - 2(v_1 v_2)^2 + \frac{9}{2}(n_{12}v_2)^2 v_2^2 \right. \right. \right. \\
 \left. \left. + 4(v_1 v_2)v_2^2 - 2v_2^4 \right) + \frac{G^2 m_2 m_2}{r_{12}^3} \left(\frac{39}{2}(n_{12}v_1)^2 - 39(n_{12}v_1)(n_{12}v_2) + \frac{17}{2}(\eta_{12}v_2)^2 - \frac{15}{4}v_1^2 - \frac{5}{2}(v_1 v_2) + \frac{5}{4}v_2^2 \right) \right. \\
 \left. + \frac{G^2 m_2^2}{r_{12}^3} \left(2(n_{12}v_1)^2 - 4(n_{12}v_1)(n_{12}v_2) - 6(n_{12}v_2)^2 - 8(v_1 v_2) + 4v_2^2 \right) \right\} n_{12} + \left\{ + \frac{G^2 m_2^2}{r_{12}^3} (-2(n_{12}v_1) - 2(n_{12}v_2)) \right. \\
 \left. + \frac{G^2 m_1 m_2}{r_{12}^3} \left(-\frac{63}{4}(n_{12}v_1) + \frac{55}{4}(n_{12}v_2) \right) + \frac{Gm_2}{r_{12}^2} \left(-6(n_{12}v_1)(n_{12}v_2)^2 + \frac{9}{2}(n_{12}v_2)^3 + (n_{12}v_2)v_1^2 - 4(n_{12}v_1)(v_1 v_2) \right. \right. \\
 \left. \left. + 4(n_{12}v_2)(v_1 v_2) + 4(n_{12}v_1)v_2^2 - 5(n_{12}v_2)v_2^2 \right) \right\} v_{12} \right]
 \end{aligned} \tag{7}$$

$$a_1^{2.5PN} = \frac{1}{c^5} \left[\left\{ \frac{208G^3 m_1 m_2^2}{15r_{12}^4} (n_{12}v_1 v_2) - \frac{24G^3 m_1^2 m_2}{5r_{12}^4} (n_{12}v_1 v_2) + \frac{12G^2 m_1 m_2}{5r_{12}^3} (n_{12}v_1 v_2) v_{12}^2 \right\} n_{12} + \left\{ \frac{8G^3 m_1^2 m_2}{5r_{12}^4} - \frac{32G^3 m_1 m_2^2}{5r_{12}^4} - \frac{4G^2 m_1 m_2}{5r_{12}^3} v_{12}^2 \right\} \right] \tag{8}$$

Calculating GW strain from inner binary

GW strain is calculated using the equations provided by (Hinder et al., 2010), given by

$$h_+ = -\frac{M\eta}{R} \left\{ (\cos^2 \iota + 1) \left[\left(-\dot{r}^2 + r^2 \dot{\phi}^2 + \frac{M}{r} \right) \cos 2\Phi + 2r\dot{r}\dot{\phi} \sin 2\Phi \right] + \left(-\dot{r}^2 - r^2 \dot{\phi}^2 + \frac{M}{r} \right) \sin^2 \iota \right\}. \quad (9)$$

$$h_\times = -\frac{2M\eta}{R} \cos \iota \left\{ \left(-\dot{r}^2 + r^2 \dot{\phi}^2 + \frac{M}{r} \right) \sin 2\Phi - 2r\dot{r}\dot{\phi} \cos 2\Phi \right\}. \quad (10)$$

where M is total mass, η is symmetric mass ratio, R is distance of source to observer, r is relative separation vector, ϕ is orbital phase and $\Phi = (\phi - \chi)$ where (χ, ι) are the observer angles.

Testing the code

We test our code by comparing our the semimajor axis decay rate from our simulation to the theoretical decay rate (Peters and Mathews, 1963),

$$\frac{da}{dt} = -\frac{64}{5} \frac{G^3 m_1 m_2 (m_1 + m_2)}{a^3 (1 - e^2)^{7/2}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (11)$$

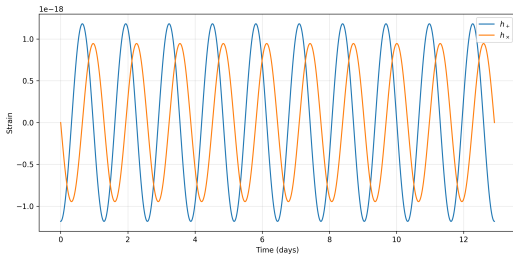


Figure 3: Strain h_+ , h_x from circular binary

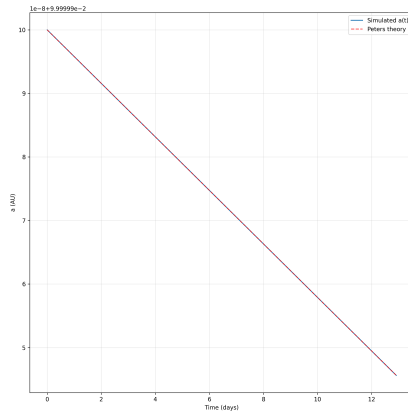


Figure 4: Orbital decay: Simulation vs Peters

Strain from a KL active inner binary

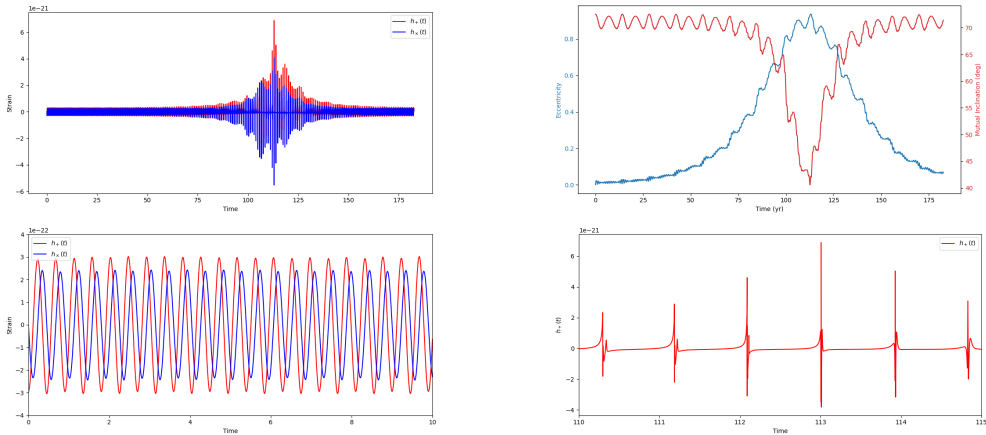


Figure 5: *Top left*: Strain, *Top right*: eccentricity-inclination cycle, *Bottom left*: Strain of binary of first 10 days and *Bottom right*: Peak strain of a hierarchical triple system $m_1 = 10M_{\odot}$, $m_2 = 10M_{\odot}$, $m_3 = 10^3M_{\odot}$, $a_{in} = 0.05AU$, $a_{out} = 1AU$ observed at a distance of $R = 8kpc$ and an observer angle of $(\chi, \iota) = (0^{\circ}, 60^{\circ})$

Frequency Domain Signal

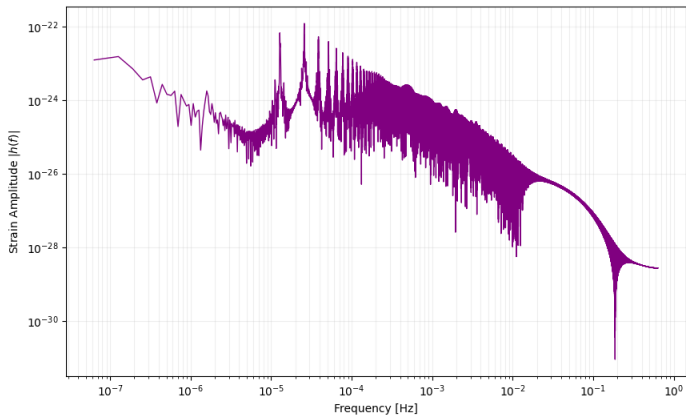


Figure 5: GW signal of hierarchical triple system $m_1 = 10M_\odot$, $m_2 = 10M_\odot$, $m_3 = 10^3 M_\odot$, $a_{in} = 0.05AU$, $a_{out} = 1AU$ observed at a distance of $R = 8kpc$ and an observer angle of $(\chi, \iota) = (0^\circ, 60^\circ)$ in frequency domain

References I

- Blaes, O., Lee, M. H., and Socrates, A. (2002). The kozai mechanism and the evolution of binary supermassive black holes. *The Astrophysical Journal*, 578(2):775.
- Blanchet, L. (2014). Gravitational Radiation from Post-Newtonian Sources and Inspiralling Compact Binaries. *Living Reviews in Relativity*, 17(1):2.
- Einstein, A. (1918). Über Gravitationswellen. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 154–167.
- Gupta, P., Suzuki, H., Okawa, H., and Maeda, K.-i. (2020). Gravitational waves from hierarchical triple systems with kozai-lidov oscillation. *Phys. Rev. D*, 101:104053.
- Hinder, I., Herrmann, F., Laguna, P., and Shoemaker, D. (2010). Comparisons of eccentric binary black hole simulations with post-newtonian models. *Phys. Rev. D*, 82:024033.
- Kozai, Y. (1962). Secular perturbations of asteroids with high inclination and eccentricity. *Astrophysical Journal*, 67:591–598.
- Lidov, M. (1962). The evolution of orbits of artificial satellites of planets under the action of gravitational perturbations of external bodies. *Planetary and Space Science*, 9(10):719–759.
- Naoz, S. (2016). The eccentric kozai-lidov effect and its applications. *Annual Review of Astronomy and Astrophysics*, 54(1):441–489.

References II

- Peters, P. C. and Mathews, J. (1963). Gravitational radiation from point masses in a keplerian orbit. *Phys. Rev.*, 131:435–440.
- Pfister, H. (2007). On the history of the so-called Lense-Thirring effect. *General Relativity and Gravitation*, 39(11):1735–1748.
- Taylor, J. H. and Weisberg, J. M. (1982). A new test of general relativity - Gravitational radiation and the binary pulsar PSR 1913+16. , 253:908–920.